

Jenni Mae Wellness

Raising Your Ferritin

The Complete Guide

A three-layer approach to gut health, mineral balance, liver support, and dietary iron absorption, to discuss with your healthcare provider

Before You Begin

This guide is educational and general in nature. It is not personalised medical advice, and it does not diagnose, treat, or replace assessment by a qualified healthcare provider.

Do not start any new supplement, or change an existing one, without first speaking to your doctor or pharmacist, including about the right dose for you, given your health history and current medications.

This guide is not appropriate for everyone. It should not be used by anyone with haemochromatosis or another iron-overload condition, anyone on iron-chelation therapy, or anyone who has not had recent bloodwork confirming low iron stores. Raising iron intake when it is not needed can be harmful.

If you are pregnant, breastfeeding, managing a chronic illness, or taking prescription medication, talk to your healthcare provider before using any part of this guide.

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Know Your Number

Ferritin Level Reference (for context only)

Ferritin is measured in micrograms per litre ($\mu\text{g/L}$). The bands below are used by some practitioners to think about iron reserves beyond the standard lab reference range. They are illustrative, not a diagnostic tool, and they are not a substitute for your doctor's interpretation of your bloodwork in the context of your overall health.

Band	Range ($\mu\text{g/L}$)	General description
A+	150–200	High reserves. Some practitioners associate this range with strong buffer for tissue repair and higher training loads, though optimal ranges are debated and vary by individual.
A/B	100–149	Reserves generally considered adequate for typical daily demands.
C	50–99	Often described as a common range where fatigue can occur despite "normal" bloodwork, worth discussing with your doctor.
D	30–49	Lower reserves; symptoms such as hair changes or fatigue are more commonly reported. Worth medical follow-up.
F	15–29	Low reserves. Medical review is recommended rather than self-management.

Standard lab reference ranges for ferritin (commonly around 12–150 $\mu\text{g/L}$, varying by lab) are population averages. Some people notice symptoms within what a lab reports as "normal." This is a reasonable thing to raise with your doctor, it isn't a basis for self-diagnosis.

Introduction

Why Your Ferritin May Be Low

Ferritin is the protein your body uses to store iron. When it's low, common symptoms include fatigue, hair shedding, brain fog, feeling the cold more, and breathlessness with exertion. A standard first response is iron supplementation and more iron-rich food. For some people this is enough; for others, ferritin doesn't move much even with supplementation, and it's worth understanding why before increasing intake further.

Low ferritin isn't always a single-variable problem. Gut absorption, mineral cofactors, and liver/inflammation status can all influence how much iron you actually retain and store. This guide walks through those three layers at a general, educational level, alongside information on iron-rich foods and absorption. Any changes to supplementation (what to take and how much) should be worked out with your healthcare provider, ideally alongside bloodwork.

Part One: Fix Your Gut First

Dietary iron has to be absorbed through the gut, so gut health is a reasonable place to start when ferritin isn't responding to diet alone.

Stomach Acid

Dietary iron exists as ferric iron (Fe^{3+}) and ferrous iron (Fe^{2+}); intestinal cells absorb the ferrous form. Stomach acid helps convert ferric to ferrous iron, and also helps break iron out of the food matrix it's bound up in, so the mineral is available for absorption in the first place. Low stomach acid (hypochlorhydria) is thought to be common and can be worsened by chronic stress, long-term antacid or PPI use, age, frequent snacking rather than set meals, drinking large amounts of fluid with meals, and eating in a rush or under stress.

Signs Some People Associate with Low Stomach Acid

Commonly reported signs include bloating or burping soon after meals, a feeling of fullness or heaviness from small meals, gas, undigested food visible in stools, brittle nails, and fatigue. Reflux or heartburn are also sometimes reported, this can seem counterintuitive, but low acid is one of several proposed contributors some practitioners discuss alongside reflux, among other causes. These signs are non-specific and overlap with many other digestive conditions, so they're a reason to raise the topic with your doctor rather than to self-diagnose.

Habits That May Support Stomach Acid

A few eating habits are commonly discussed as ways to support your body's own stomach acid production, alongside (not instead of) addressing any underlying medical cause:

- Eat at set meals rather than grazing throughout the day, frequent small bouts of eating are thought to give digestive secretions less chance to build back up between meals.
- Limit drinking large amounts of fluid with meals, and drink most fluids between meals instead, large volumes of liquid with food are thought to dilute stomach acid.
- Avoid very large meals where possible, overloading the stomach at one sitting is thought to place more demand on digestive secretions than they can comfortably meet.
- Eat in a calm setting where practical, sitting down, chewing thoroughly, and avoiding stressful conversation or work at the table, the nervous system's "rest and digest" state supports normal digestive secretions, while a stressed state tends to suppress them.
- A small amount of diluted apple cider vinegar, or bitter foods (e.g. rocket, chicory), or lemon juice in a little warm water, taken shortly before a meal, are dietary approaches some people find helpful.
- A small amount of cayenne pepper or other gently warming spice with a meal is another approach some people use to encourage digestive secretions.

Betaine HCl is another supplement sometimes used to support stomach acid directly. As with any supplement in this guide, the right dose is individual, and it's worth confirming with your doctor whether it's a good fit for you, especially since reflux and bloating can have several different causes, and Betaine HCl is not appropriate for everyone (including anyone with a history of gastritis, ulcers, or who takes NSAIDs or corticosteroids regularly).

Testing Stomach Acid at Home

There's no validated at-home test for stomach acid, if you want a reliable answer, the accurate option is a test ordered by your doctor (such as the Heidelberg capsule test, where available). That said, two informal, low-risk self-checks are commonly discussed. Both are indicative at best, not diagnostic, and shouldn't be used to start or stop any medication, including PPIs or antacids, without medical advice.

- **The bicarbonate (baking soda) test:** First thing in the morning, before eating or drinking anything, mix ¼ teaspoon of bicarbonate of soda into 150-200ml of cold water and drink it. Time how long it takes to burp. The idea is that stomach acid reacting with the bicarbonate produces gas (carbon dioxide), so a burp within about 2-3 minutes is taken to suggest adequate acid, while a longer time, or no burp within 5 minutes, is taken to suggest low acid. This test is popular in wellness circles but has not been clinically validated, results can vary based on how much air is swallowed while drinking, so treat it as a rough, informal indicator rather than a result to act on alone.
- **The Betaine HCl challenge:** Some practitioners use a Betaine HCl capsule taken partway through a protein-containing meal, on the idea that a warming or burning sensation in the stomach indicates acid levels are already adequate (in which case the supplement isn't needed), while no sensation, even as the dose is gradually increased over separate meals, is taken to suggest low acid. This is more involved and carries a real risk of a burning sensation, so it should only be done under the guidance of a healthcare provider or experienced practitioner, and never if you have a history of ulcers, gastritis, or reflux, or take NSAIDs, corticosteroids, or blood thinners.

The Gut Microbiome and Iron Absorption

The gut microbiome appears to play a role in iron metabolism. Certain bacteria (e.g. Lactobacillus species) are associated with short-chain fatty acid production that supports intestinal lining integrity, which affects absorptive surface area. When this lining is compromised, absorption and inflammatory signalling can both be affected.

Fermented Foods

- Kimchi: a source of Lactobacillus strains; commonly used as a condiment alongside meals.
- Kefir (dairy or coconut): a broad-spectrum source of probiotic bacteria and yeasts.
- Sauerkraut: unpasteurised, refrigerated varieties retain live cultures; choose based on personal tolerance.

As with any fermented food, start with small amounts if you're not used to them, and reduce or stop if they cause digestive discomfort.

Lactoferrin

Lactoferrin is an iron-binding glycoprotein found naturally in breast milk, colostrum, saliva, and tears. It's often discussed as a gentler option than standard iron tablets, which can cause constipation and gut discomfort in some people. If you're considering a lactoferrin supplement, dosing is individual, so it's worth confirming what makes sense for you with your doctor.

Part Two: The Iron in Your Food

Dietary iron comes in two forms with different absorption characteristics.

Heme Iron

Heme iron is found in animal foods, red meat, organ meat, poultry, and fish. It's absorbed directly, without the conversion step non-heme iron requires. Reported absorption rates range roughly 15–35%, and tend to be higher when iron stores are already low. Heme iron absorption is not strongly affected by other foods eaten at the same meal.

Food	Serving	Iron (mg)
Beef liver	100g cooked	6.5 mg
Chicken liver	100g cooked	9.0 mg
Lamb kidney	100g cooked	5.7 mg
Beef mince (lean)	150g cooked	3.8 mg
Lamb (shoulder)	150g cooked	3.2 mg
Venison / kangaroo	150g cooked	4.5 mg
Oysters	100g	7.0 mg
Sardines (tinned)	100g	2.5 mg
Tuna (tinned)	100g	1.4 mg
Chicken breast	150g cooked	1.0 mg

Liver and other organ meats are nutrient-dense (iron, copper, vitamin A, B12, folate, choline). Note that vitamin A from animal sources (retinol) can be harmful in excess, particularly in pregnancy, if you eat liver regularly and are pregnant or trying to conceive, check appropriate serving frequency with your doctor.

Non-Heme Iron

Non-heme iron is found in plant foods, eggs, and fortified products, and must be converted from ferric to ferrous form before absorption. Reported absorption rates are lower, roughly 2–10% with adequate iron stores, potentially higher when iron-deficient. Absorption of non-heme iron is more sensitive to what else is eaten at the same meal.

Food	Serving	Iron (mg)
Lentils (cooked)	180g (1 cup)	6.6 mg
Firm tofu	150g	3.0 mg
Chickpeas (cooked)	180g (1 cup)	4.7 mg
Kidney beans	180g (1 cup)	3.9 mg
Pumpkin seeds	30g (2 tbsp)	2.5 mg
Hemp seeds	30g (3 tbsp)	2.4 mg

Food	Serving	Iron (mg)
Spinach (cooked)	180g	6.4 mg
Dark chocolate (85%+)	30g	3.4 mg
Blackstrap molasses	1 tbsp	4.0 mg
Dried apricots	50g (8 halves)	1.8 mg
Fortified breakfast cereal	45g serving	3–8 mg

Part Three: Enhance & Inhibit

What May Enhance Non-Heme Iron Absorption

Vitamin C is commonly cited as a helpful pairing with non-heme iron, as it can assist conversion to the absorbable ferrous form. Food-based vitamin C sources are a reasonable, low-risk way to do this.

Vitamin C food	Approx. vitamin C
Red capsicum (half)	105 mg
Kiwi fruit (1 medium)	92 mg
Orange juice (200ml)	70 mg
Broccoli (80g, lightly cooked)	65 mg
Strawberries (handful)	58 mg
Lemon juice (squeezed over food)	15–25 mg

- Acidic/fermented foods (lemon juice, vinegar) may help keep iron in a more absorbable form.
- Including a small amount of meat, fish, or poultry alongside plant-based iron sources is associated with improved non-heme iron absorption in some studies.
- Cooking, soaking, or choosing sprouted legumes and grains can reduce phytate content, a known inhibitor.

What May Inhibit Iron Absorption

These compounds can reduce iron absorption when eaten close to an iron-containing meal. This isn't a reason to avoid these foods generally (many are nutritious) simply a reason to consider spacing them from iron-focused meals if you're actively working on your iron status.

Inhibitor	Found in / notes
Calcium	Dairy, fortified plant milks. Some research suggests even moderate calcium intake can reduce iron absorption at the same meal.
Tannins & polyphenols	Tea (including herbal), coffee, red wine, cocoa. Commonly advised to separate from meals by roughly 1–2 hours.
Phytates (phytic acid)	Wholegrains, legumes, nuts, seeds. Reduced by soaking, sprouting, and cooking.
Oxalates	Raw spinach, beet greens, chard. Cooking reduces oxalate content.
Coffee polyphenols	Including decaf. Often grouped with tea for meal-timing purposes.
Soy protein	Fermented soy (miso, tempeh, natto) generally has lower inhibitor content than unfermented tofu/soy milk.

Part Four: The Mineral Symphony

Several minerals interact with iron metabolism. This section is general nutrition information; supplementing any of these should be discussed with your healthcare provider, particularly since combining multiple supplements without guidance can lead to interactions or excess intake.

Copper

Copper supports a liver enzyme (ceruloplasmin) involved in preparing iron for transport in the blood. Good dietary sources include liver, oysters, dark chocolate, shiitake mushrooms, cashews, sesame seeds, and spirulina. Food-based copper intake is generally a lower-risk way to support this than supplementing, since excess supplemental copper carries its own risks.

Magnesium

Magnesium is involved in energy-dependent processes relevant to red blood cell production and is a commonly under-consumed mineral. Food sources include dark leafy greens, pumpkin seeds, dark chocolate, avocado, and legumes. Magnesium glycinate and magnesium malate are forms some people find well absorbed. Dosing is individual, so it's worth confirming the right form and amount for you with your doctor, especially if you have any kidney concerns.

Zinc

Zinc and iron share an intestinal transporter and can compete for absorption. Zinc also supports gut lining integrity. If you take both iron and zinc supplements, it's generally best to take them at separate times of day. Dosing is individual, so confirm the right amount for you with your doctor.

Sodium and Potassium

These electrolytes support the cellular processes involved in nutrient transport generally. People who are very active, sweating heavily, or on medically-restricted sodium diets should talk to their doctor about electrolyte needs rather than adjusting intake independently, particularly if there's a heart, kidney, or blood pressure condition involved.

Mineral	Role (general)	Food sources
Copper	Supports iron transport enzyme (ceruloplasmin)	Liver, oysters, cashews, dark chocolate
Magnesium	Supports red blood cell production, cellular energy	Pumpkin seeds, greens, dark chocolate
Zinc	Gut lining support; shares absorption pathway with iron	Oysters, beef, pumpkin seeds, hemp
Sodium	General cellular transport processes	Sea salt, olives, fermented vegetables
Potassium	Works alongside sodium in cellular transport	Avocado, banana, sweet potato, beans

Part Five: Support Your Liver & Lower Inflammation

The liver plays a central role in recycling and regulating iron storage, via a hormone called hepcidin. Chronic inflammation is associated with elevated hepcidin, which can suppress iron availability even when total body iron isn't low, sometimes described as "anaemia of chronic inflammation." Because ferritin itself can rise during inflammation (it's an acute-phase protein), your doctor may check CRP (a marker of inflammation) alongside ferritin to get a clearer picture. This interpretation is best done by a healthcare provider, since high or low ferritin can mean different things depending on the wider clinical picture.

Signs Some People Associate with Sluggish Liver Function

Morning fatigue, waking overnight, difficulty digesting fats, pale stools, persistent skin issues, feeling worse after alcohol or rich food, and severe PMS are sometimes discussed anecdotally in this context. These are non-specific and can have many causes, they're worth raising with your doctor rather than self-treating.

Liver-Supportive Habits & Supplements

General liver-supportive habits include eating adequate protein and choline-rich foods (e.g. eggs), moderating alcohol, maintaining stable blood sugar, prioritising sleep, and managing chronic stress.

A few supplements are commonly discussed alongside liver support and iron recovery:

- Milk Thistle (Silymarin): One of the most researched liver-supportive herbs, valued for its antioxidant properties and role in supporting the liver's detox pathways.
- N-Acetylcysteine (NAC): A precursor to glutathione, the liver's primary antioxidant. Often used to support the liver's capacity to manage oxidative stress.
- Dandelion Root & Artichoke Extract: Traditionally used bitters that stimulate bile production, supporting digestion and fat-soluble vitamin absorption.

Dosing for these varies from person to person, and depends on your health history and any medications you take, so it's worth confirming the right choice and amount with your doctor or pharmacist before starting.

Reducing Inflammation: General, Low-Risk Habits

Habit	Why it's relevant
Omega-3s (oily fish, algae oil, or fish oil supplement)	Commonly associated with lower inflammatory markers. If choosing a supplement, dosing is individual, so confirm the right amount for you with your doctor, especially if you take blood thinners.
Turmeric/curcumin (food or supplement, with black pepper for absorption)	Traditionally used and generally well tolerated as a food. Concentrated curcumin supplements have their own dosing considerations, so check with your pharmacist first.
Stable blood sugar	Prioritising protein and fibre, and limiting refined carbohydrates, is a generally low-risk habit associated with lower inflammatory markers.
Consistent sleep	Sleep quality is linked to inflammatory and hormonal regulation.
Stress management	Practices such as slow breathing, time outdoors, or gentle movement are low-risk and widely recommended.
Reducing alcohol	Alcohol affects liver function broadly; reducing intake is a generally safe recommendation for most adults.

Protocol Summary

The list below is a starting framework for a conversation with your healthcare provider, not a prescription. Specific supplements and doses should be individualised based on your bloodwork, health history, and current medications.

Morning

- Consider a breakfast with a protein source and a vitamin-C-rich food (e.g. kiwi, berries, citrus) alongside any iron-rich foods.
- If you're taking supplements like lactoferrin or magnesium, generally take them with food, and at the dose you and your doctor have agreed on.
- Consider leaving a gap between tea/coffee and iron-focused meals.

Midday Meal

- Iron-rich foods a few times a week if suitable for you: red meat, organ meat, shellfish, or plant sources such as lentils and legumes.
- Pair plant-based iron with a vitamin-C source (e.g. lemon over lentils, capsicum with bean dishes).
- A fermented vegetable side (kimchi or sauerkraut), if well tolerated.
- Consider spacing dairy, tea, or coffee away from iron-focused meals.

Throughout the Day

- Kefir, if well tolerated, away from main iron-containing meals.
- Stay hydrated; ask your doctor about electrolyte needs if you're very active.
- Any supplements should follow your healthcare provider's guidance on product, dose, and timing.

Evening

- Prioritise consistent, adequate sleep.
- Moderate alcohol intake, particularly while focused on improving iron status.

Testing & Follow-Up

Ask your doctor whether a full iron panel is appropriate, this may include serum iron, transferrin saturation, TIBC, and CRP alongside ferritin. Ferritin alone doesn't tell the whole story, and your doctor is best placed to interpret your results in the context of your symptoms and history, and to advise on any supplementation and appropriate doses.

Who Should Not Use This Guide Without Medical Supervision

Anyone with haemochromatosis or another iron-overload condition.

Anyone on iron-chelation therapy or with a history of iron toxicity.

Anyone who has not had recent bloodwork confirming genuinely low ferritin/iron stores.

Anyone who is pregnant, breastfeeding, managing a chronic illness, or taking prescription medication, supplements discussed in this guide can interact with medications or be inappropriate in these situations.

Children and adolescents, this guide is written with adult women in mind and is not intended for use with minors.

This guide is for general educational purposes only. It is not medical advice and is not a substitute for individualised care from a qualified healthcare provider. Talk to your doctor before making changes to your diet, and always before starting, stopping, or changing any supplement.